Machine Learning and Deep Learning
Lecture-07

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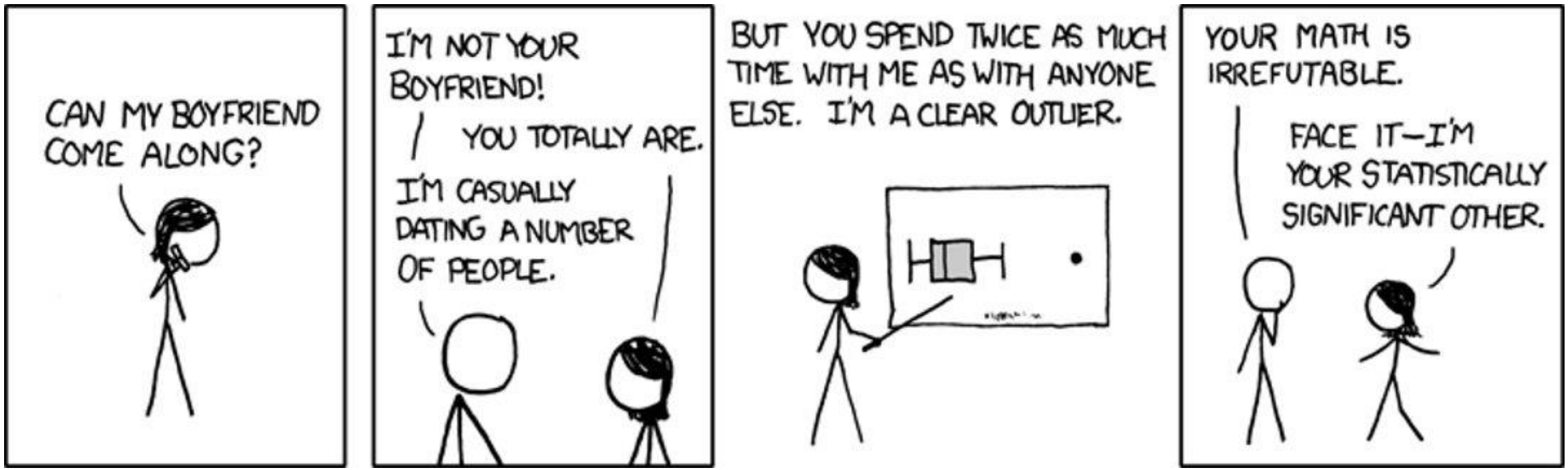
Overview

- Outlier Detection
- Finding Similar Items
- Recommender Systems
- Class Imbalance

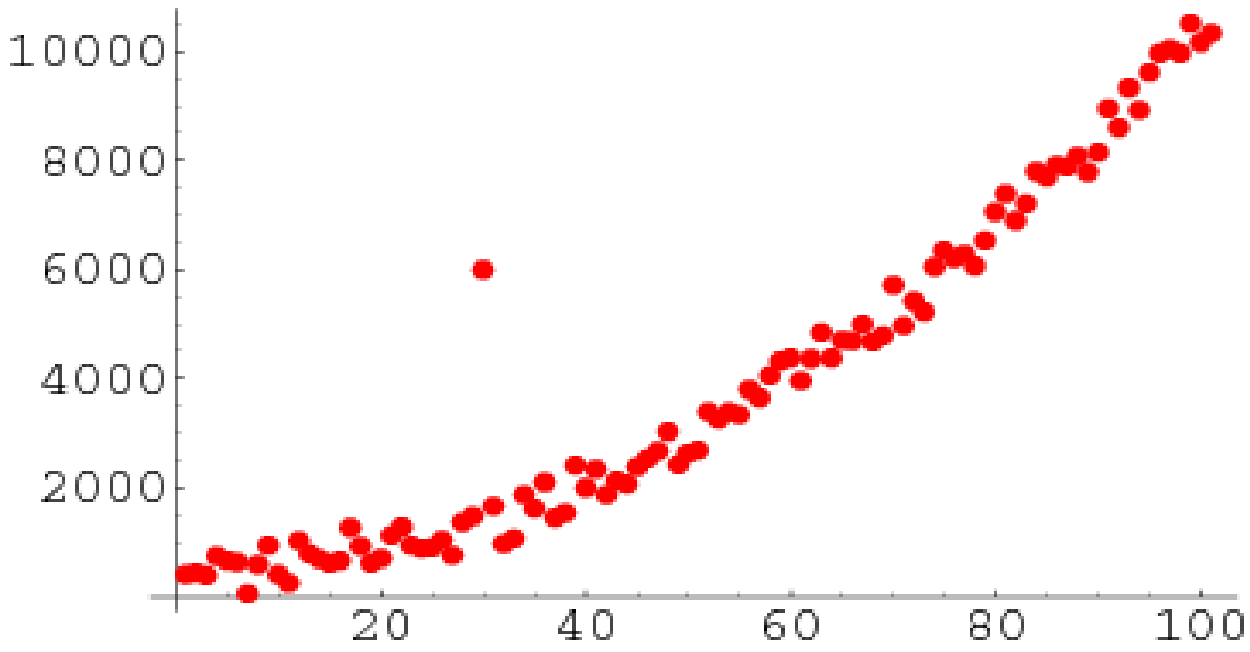
Outlier Detection



https://xkcd.com/539/

Outlier/Anomaly Detection

- Outlier: A data object that deviates significantly from the majority of observations
- Indicating that it was generated by a different underlying mechanism.
- Goal: Identify observations that are unusually different from the rest of the dataset.
- Challenge: Notion of “unusual” is difficult to define precisely and depends on the data distribution, dimensionality, and application context.

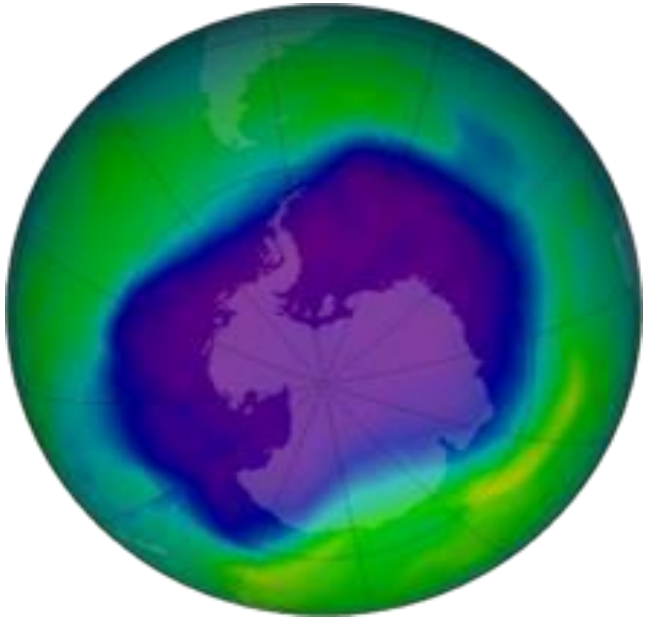


- Sources:
  - Measurement errors
  - Data entry errors
  - Data contamination from heterogeneous sources
  - Genuine rare events

http://mathworld.wolfram.com/Outlier.html

Example: Finding Holes in Ozone Layer

- Huge Antarctic ozone hole was “discovered” in 1985.
- It had been in satellite data since 1976
  - But it was flagged and filtered out by quality-control algorithm.



https://en.wikipedia.org/wiki/Ozone_depletion

Outlier Detection

- Application Domain:
  - Data cleaning
  - Security and fault detection
  - Fraud detection
  - Detecting natural disasters
  - Astronomy
  - Genetics

- Methods to detect outliers:
  - Model-based
  - Graphical approaches
  - Cluster-based
  - Distance-based
  - Supervised-learning

Global vs. Local Outliers

- Is the red point an outlier?

Global vs. Local Outliers

- Is the red point an outlier? What if we add the blue points?

Global vs. Local Outliers

- Is the red point an outlier? What if we add the blue points?

- Red point has the lowest z-score*.
  - First case: “global” outlier.
  - Second case: “local” outlier:
    - Within normal data range, but far from other points.

Z-score describes a value's relationship to the mean of a group of values. It is measured in terms of standard deviations from the mean.

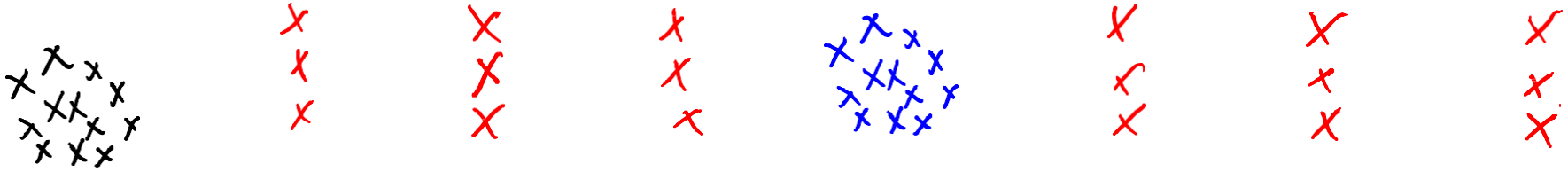
Global vs. Local Outliers

- Is the red point an outlier? What if we add the blue points?

- Red point has the lowest z-score.
  - First case : “global” outlier.
  - Second case : “local” outlier:
    - Within normal data range, but far from other points.
- Can we have outlier groups?

Global vs. Local Outliers

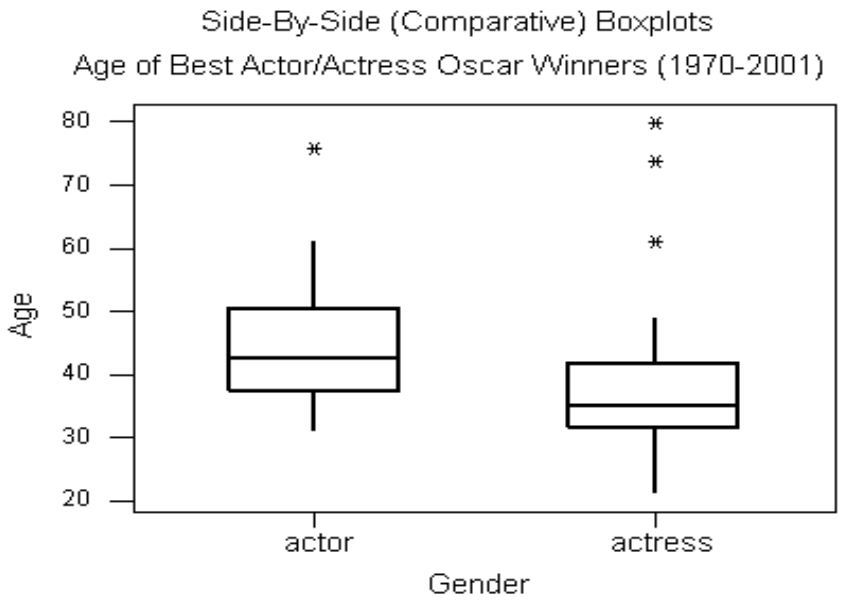
- Is the red point an outlier? What if we add the blue points?

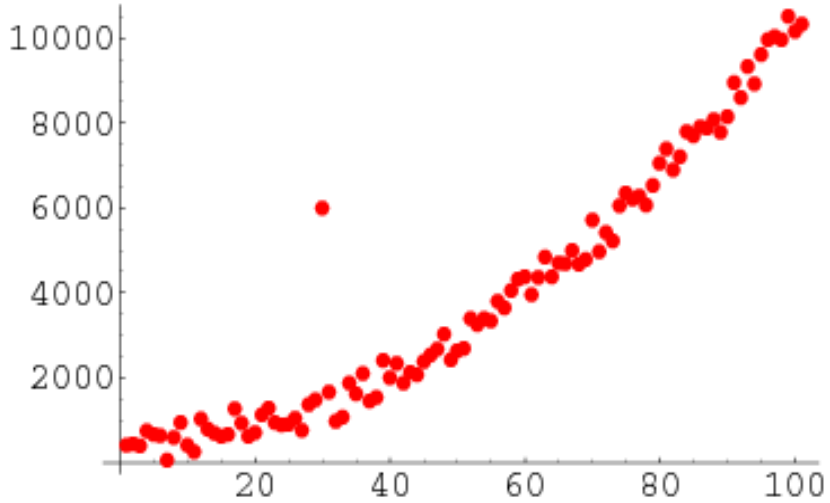


- Red point has the lowest z-score.
  - First case : “global” outlier.
  - Second case : “local” outlier:
    - Within normal data range, but far from other points.
- Can we have outlier groups?
- What about repeating patterns?

Graphical Outlier Detection

- Graphical approach to outlier detection:
  - Plot the data and look for weird points.
  - Human decides if data is an outlier.





- boxplot(x) creates a box plot of data in x.
- If x is a vector--> box.
- If x is a matrix --> one box/column of x.

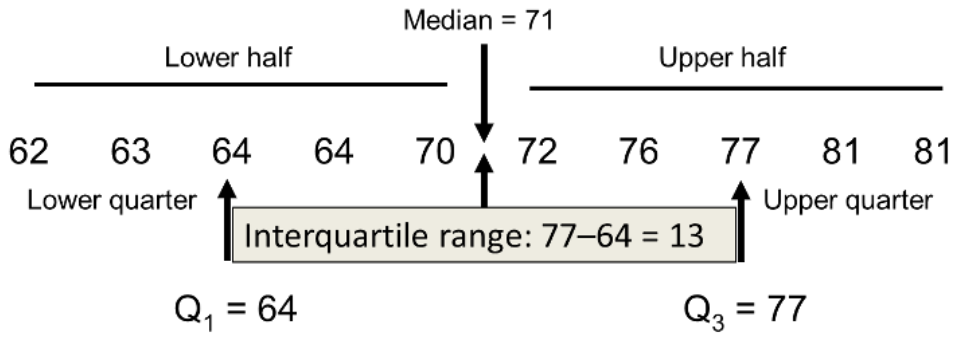
- Scatterplot detects complex patterns.
- Only 2 variables at a time.

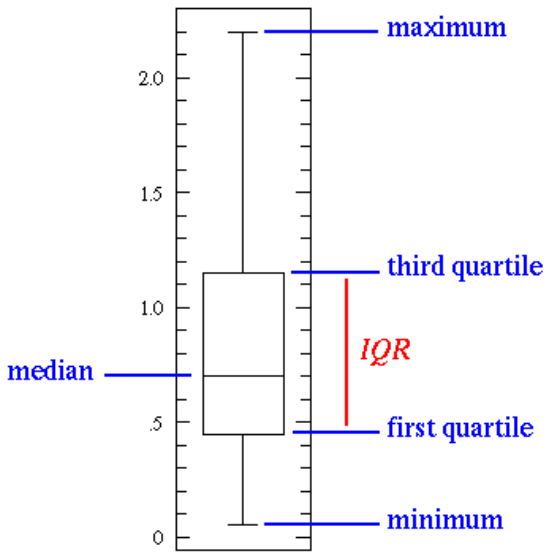
http://bolt.mph.ufl.edu/6050-6052/unit-1/one-quantitative-variable-introduction/boxplot/

http://mathworld.wolfram.com/Outlier.html

Quartiles and Boxplot

- First quartile (Q1): Median of lower half of data set.
  - 25% of numbers in data set lie below Q1 and about 75% lie above Q1.
- Third quartile (Q3): Median of upper half of data set.
  - 75% of numbers in data set lie below Q3 and about 25% lie above Q3.
- Interquartile Range (IQR): When a data set has outliers, variability is often summarized by a statistic called the interquartile range, i.e., Q3-Q1.





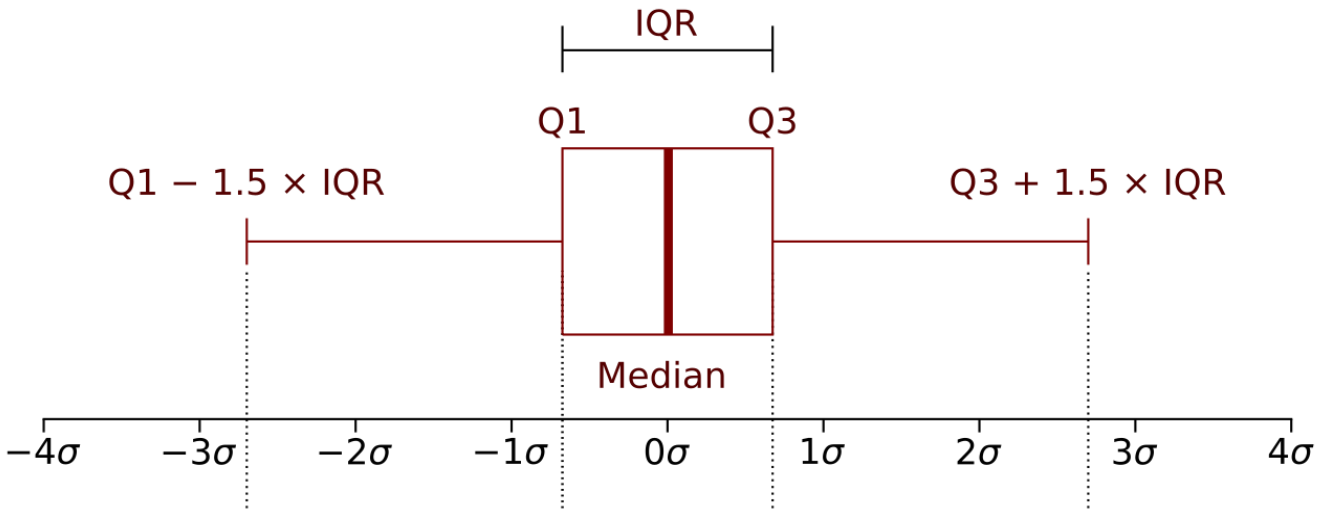
http://www.physics.csbsju.edu/stats/box2.html

- Boxplot: displays five-number summary of a set of data.
- They are minimum, Q1, median, Q3, and maximum.

http://sphweb.bumc.bu.edu/otlt/mph-modules/bs/bs704_summarizingdata/bs704_summarizingdata7.html

Box Plot: 1.5 IQR rule

- Why do we multiply the interquartile range by 1.5 to find the outlier?
- **Short answer:**
  - 1.5 IQR rule: is a rule of thumb used for identifying outliers.
  - It designates outliers based on an upper and lower boundary or “fence” outside of Q1 and Q3.
  - Anything above Q3 +1.5 x IQR is an outlier
  - Anything below Q1- 1.5xIQR is an outlier



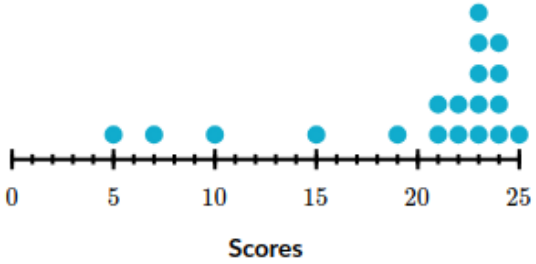
Long Answer

Identifying outliers with the 1.5xIQR rule

Google Classroom

An outlier is a data point that lies outside the overall pattern in a distribution.

The distribution below shows the scores on a driver's test for 19 applicants. How many outliers do you see?

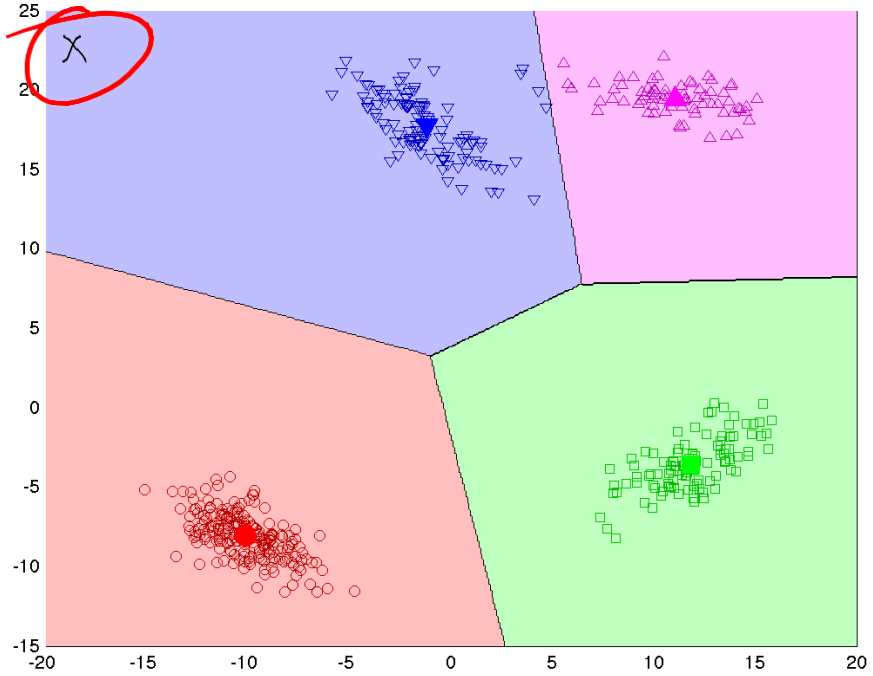


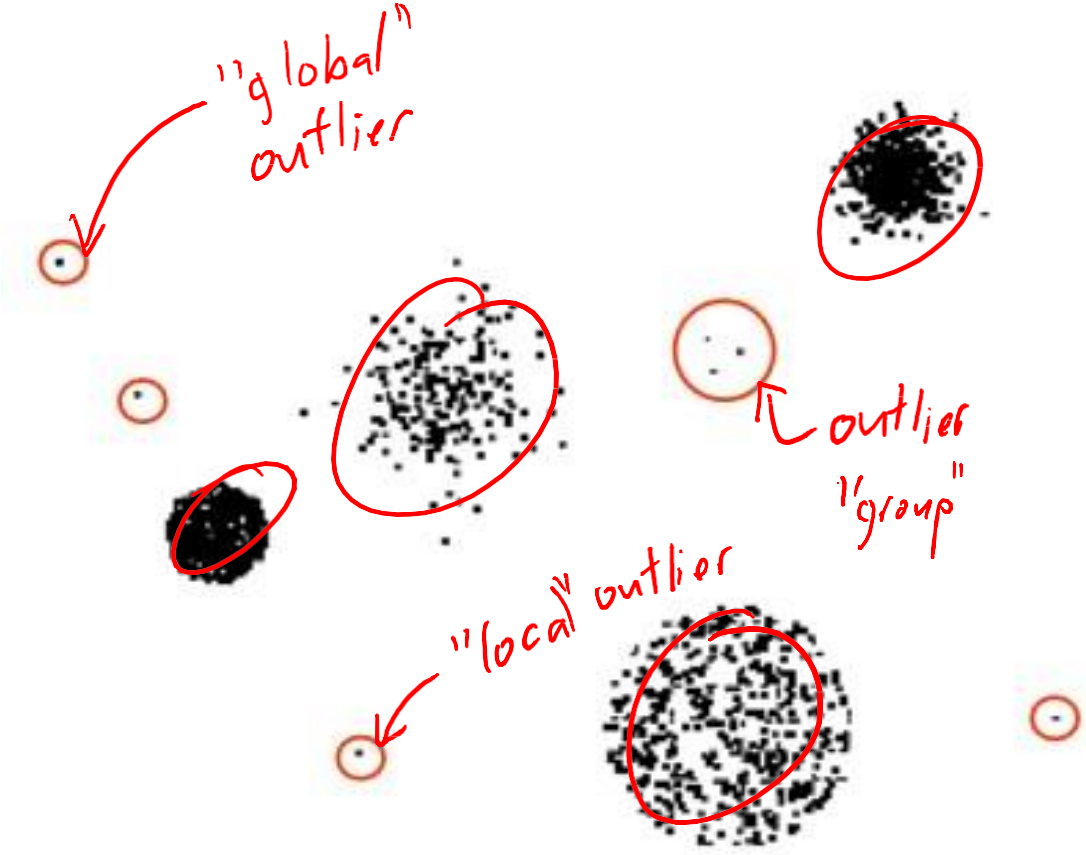
Some people may say there are 5 outliers, but someone else might disagree and say there are 3 or 4 outliers. Statisticians have developed many ways to identify what should and shouldn't be called an outlier.

A commonly used rule says that a data point is an outlier if it is more than $1.5 \cdot \text{IQR}$ above the third quartile or below the first quartile. Said differently, low outliers are below $Q_1 - 1.5 \cdot \text{IQR}$ and high outliers are above $Q_3 + 1.5 \cdot \text{IQR}$.

Cluster-Based Outlier Detection

- Detect outliers based on clustering
  - Cluster the data.
  - Find points that don't belong to clusters.





- K-means:
  - Find points that are far away from any mean.
  - Find clusters with a small number of points.

- Density-based clustering:
  - Outliers are points not assigned to cluster.

https://www-users.cs.umn.edu/~kumar001/dmbook/dmslides/chap10_anomaly_detection.pdf

Distance-Based Outlier Detection

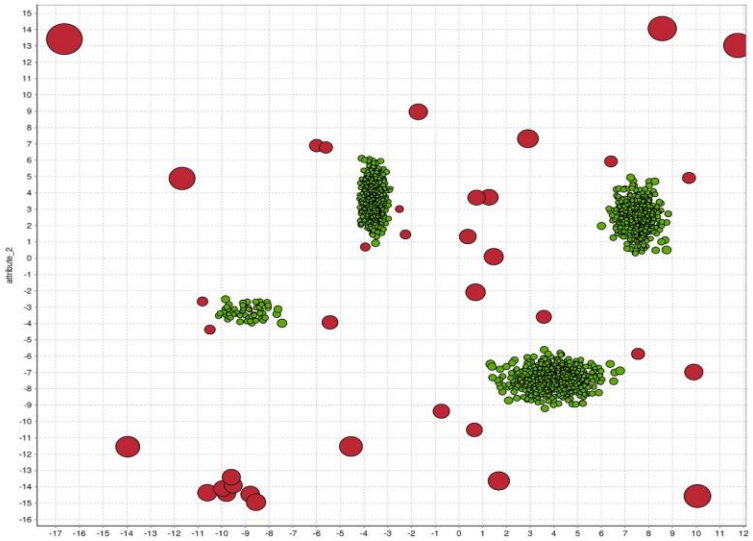
- Many outlier detection methods rely on distance measures.
- Instead of building explicit models, visualizations, or clusters, we can directly evaluate distances between data points.
- Key questions:
  - How many points lie within a radius ε?
  - What is the distance to the k-th nearest neighbor?
- Application:
  - KNN-based methods are particularly effective for detecting global outliers.
- Empirical comparisons (e.g., studies evaluating 19 methods across 10 datasets) show that:
  - Global outliers are well detected using standard KNN approaches.
  - Local outliers are better identified using local distance-based methods.

*http://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0152173

KNN-Based Outlier Detection

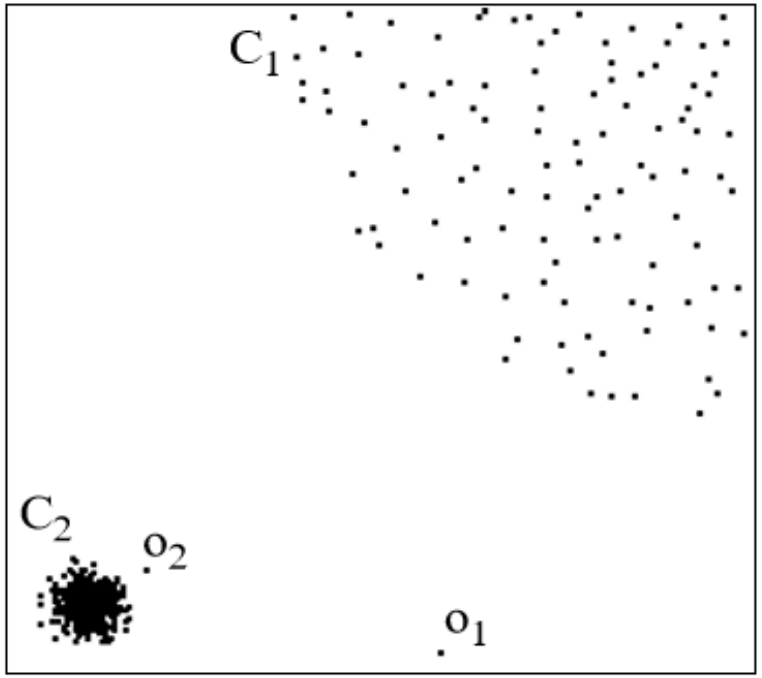
- KNN-Based Outlier Detection:
  1. For each data point, compute the average distance to its k nearest neighbors (KNN).
  2. Obtain the set of n average distances (one per data point).
  3. Rank the points in descending order based on these distances.
  4. Points with the largest average distances are considered outliers.

- Points that are far from their nearest neighbors are flagged as potential outliers.



Local Distance-Based Outlier Detection

- As with density-based clustering, problem with differing densities
- Outlier o_2 has similar density as elements of cluster C1.
- Basic idea behind local distance-based methods:
  - Outlier o_2 is “relatively” far compared to its neighbors.



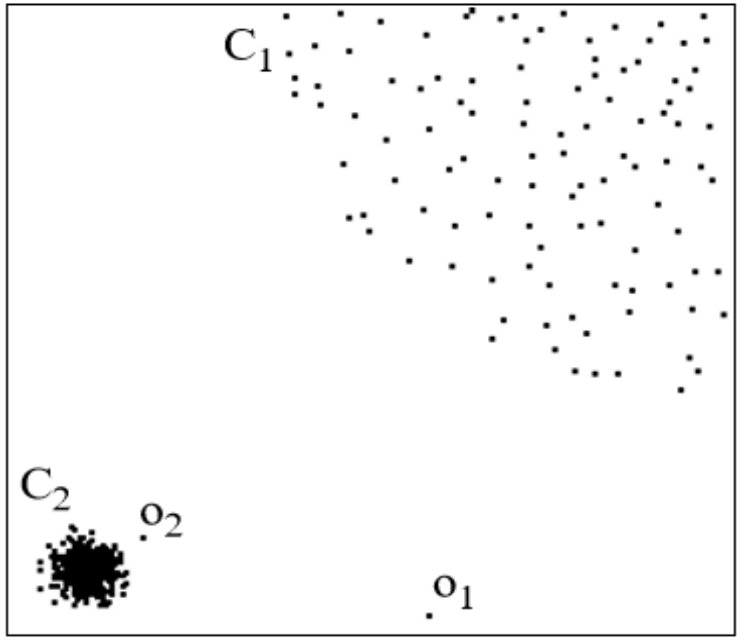
http://www.dbs.ifi.lmu.de/Publikationen/Papers/LOF.pdf

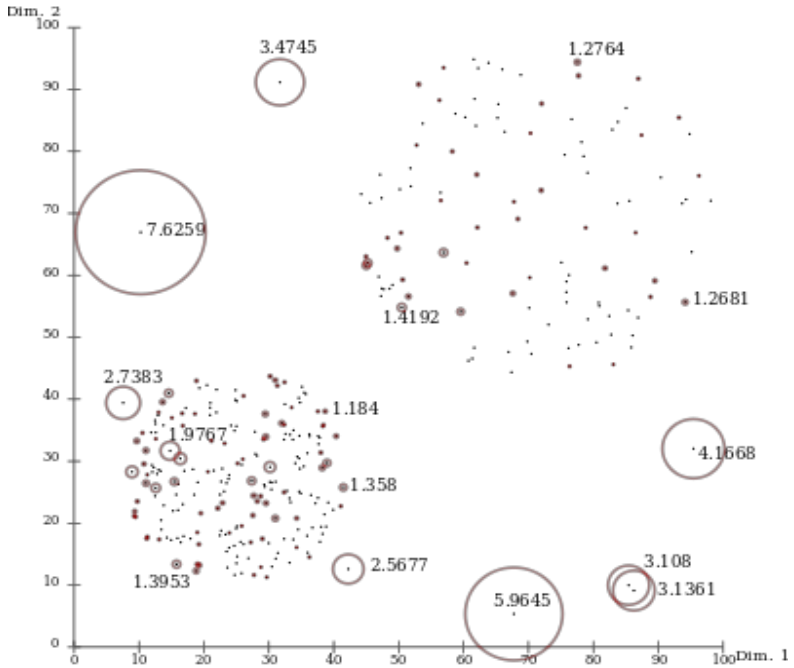
Outlierness

Outlierness Ratio for a Data Point i:

$$\frac{\text{Average distance of } i \text{ to its } k \text{ nearest neighbors}}{\text{Average distance of } i\text{'s neighbors to their own } k \text{ nearest neighbors}}$$

- If the outlierness ratio > 1, point i lies far from its neighbors than they lie from each other.
- Larger the ratio, the stronger the indication that i is a local outlier.
- A value close to 1 suggests the point has similar density to its neighborhood.

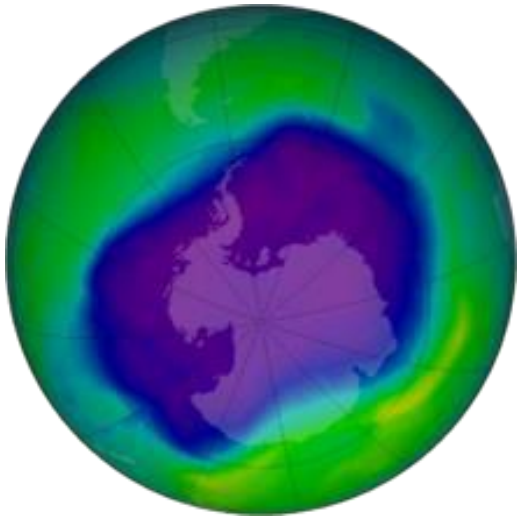




http://www.dbs.ifi.lmu.de/Publikationen/Papers/LOF.pdf
https://en.wikipedia.org/wiki/Local_outlier_factor

Problem with Unsupervised Outlier Detection

- Why wasn’t the hole in the ozone layer discovered for 9 years?
- Can be hard to decide when to report an outlier.
  - If you report too many non-outliers, users will turn you off.



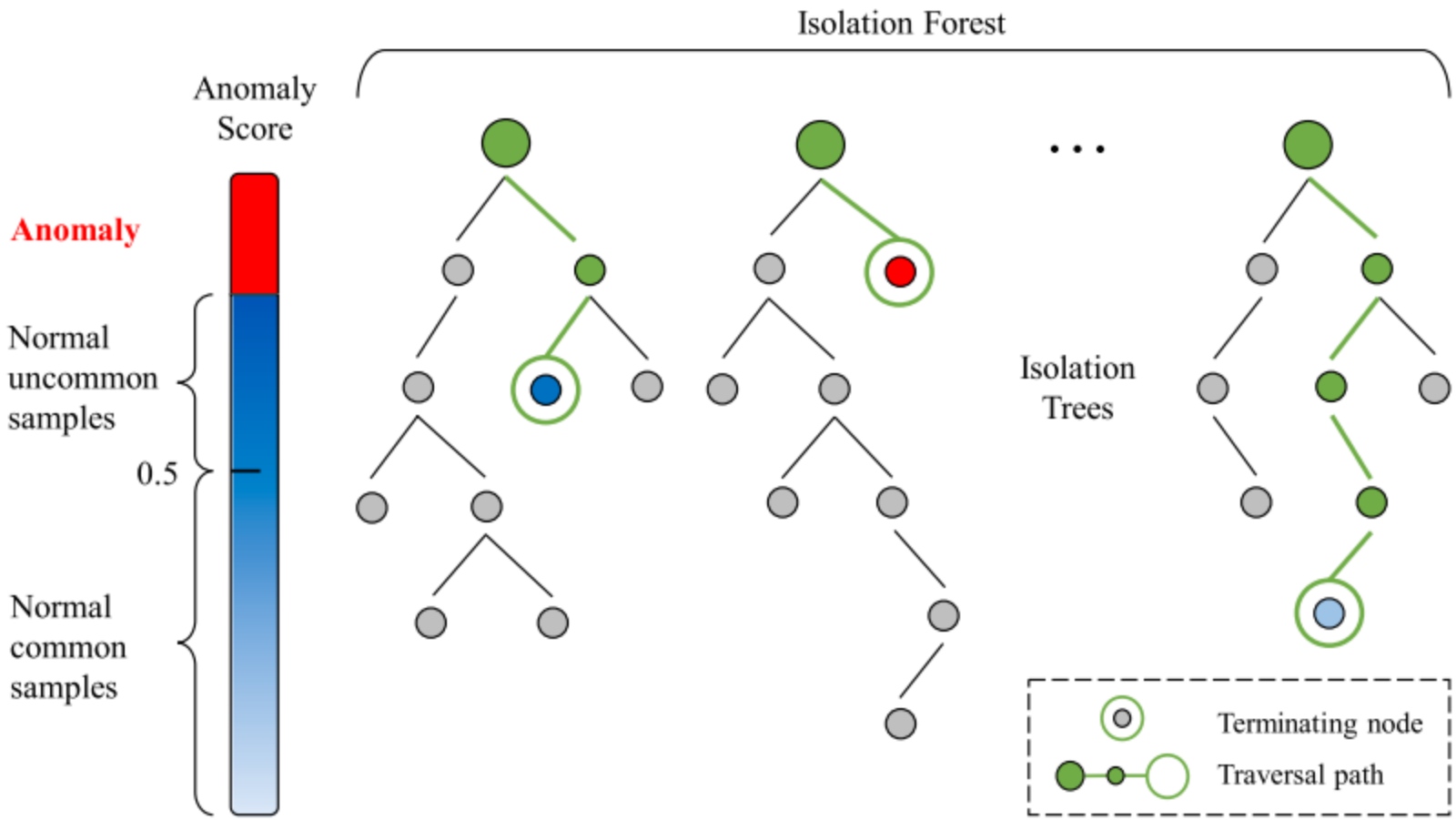
https://en.wikipedia.org/wiki/Ozone_depletion

Isolation Forests*

- A tree-based anomaly detection method.
  - Builds an ensemble of randomly generated isolation trees to compute anomaly scores for each instance.
  - Has linear time complexity (approximately $O(n)$).
  - unsupervised and scalable method.
- Performs well on large and high-dimensional datasets.
- Less stable on very small datasets (due to limited partition diversity).
- Isolates points via random recursive partitioning.
- Does not assume any specific feature distribution.
- Provides anomaly scores but limited interpretability regarding why a point is anomalous.

*Liu, Fei Tony, Ting, Kai Ming and Zhou, Zhi-Hua. "Isolation forest." Data Mining, 2008. ICDM'08. Eighth IEEE International Conference on.

Example



https://doi.org/10.1016/j.aei.2020.101139

Isolation Forests

- Randomly select a feature from the feature set.
- Randomly select a split value within the range of that feature (between min and max values among the samples in the current node).
- Partition the data into two child nodes according to the selected split value.
- Recursively repeat the process for each child node until one of the following stopping conditions is met:
  1. The node contains only one instance, or
  2. The maximum tree depth is reached, or
  3. All instances in the node have identical values.

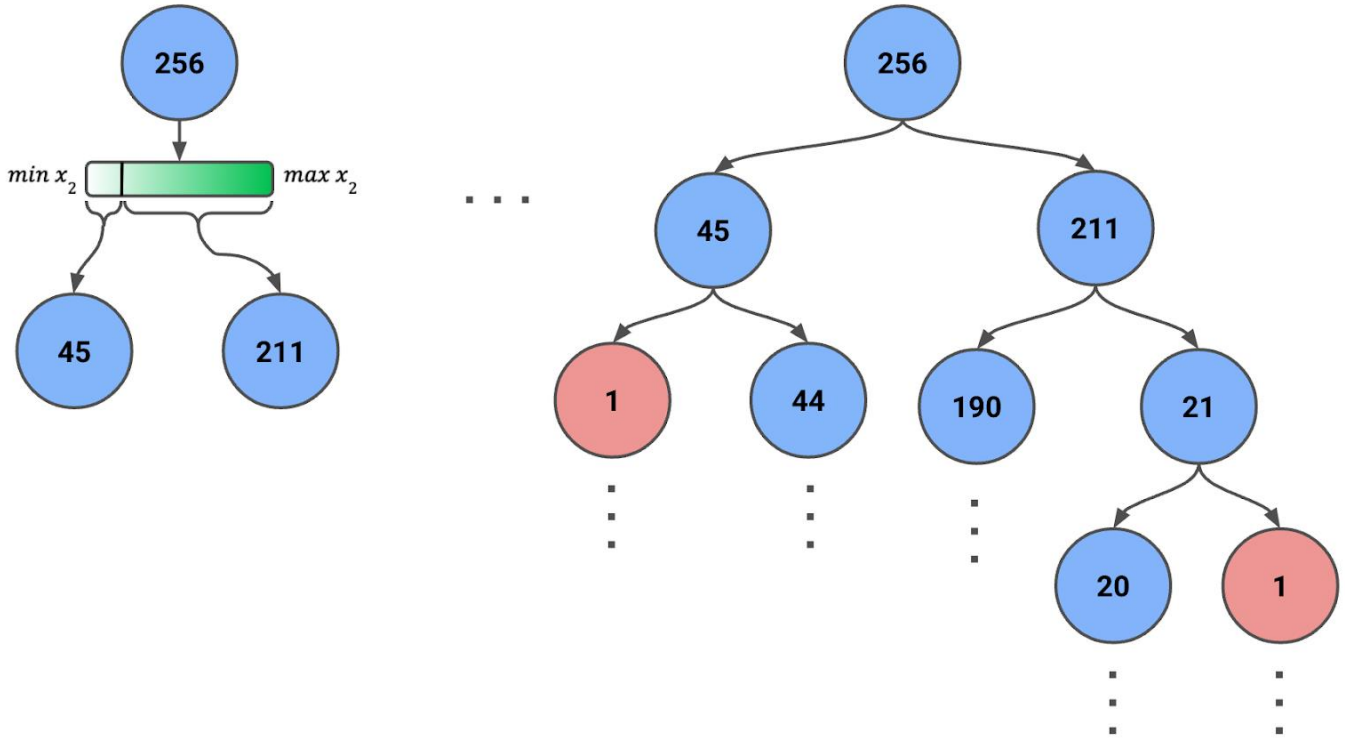
Here, we use feature's minimum and maximum values for the node's instances.

Score range: 0 (not an anomaly) and 1 (an anomaly)

Src Datacamp https://www.datacamp.com/tutorial/isolation-forest

Isolation Forests

- Topic: Fraudulent transactions detection
- Instances: 256 transactions
- Var: amount (x1) and time of day (x2)
- Note: Feature's range will change for each step.



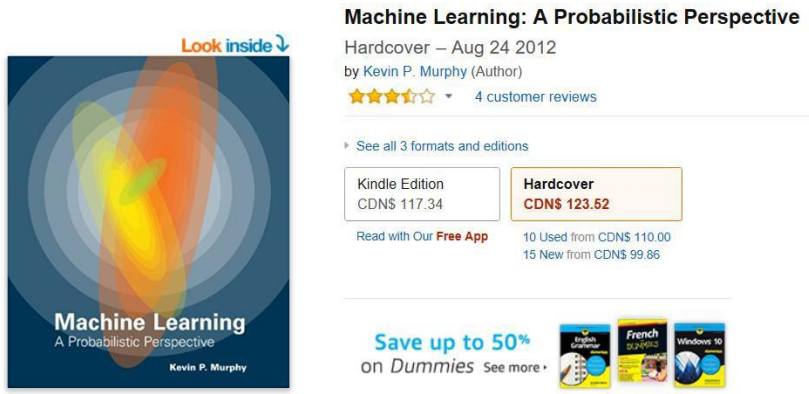
Score range: 0 (not an anomaly) and 1 (an anomaly)

Src Datacamp https://www.datacamp.com/tutorial/isolation-forest

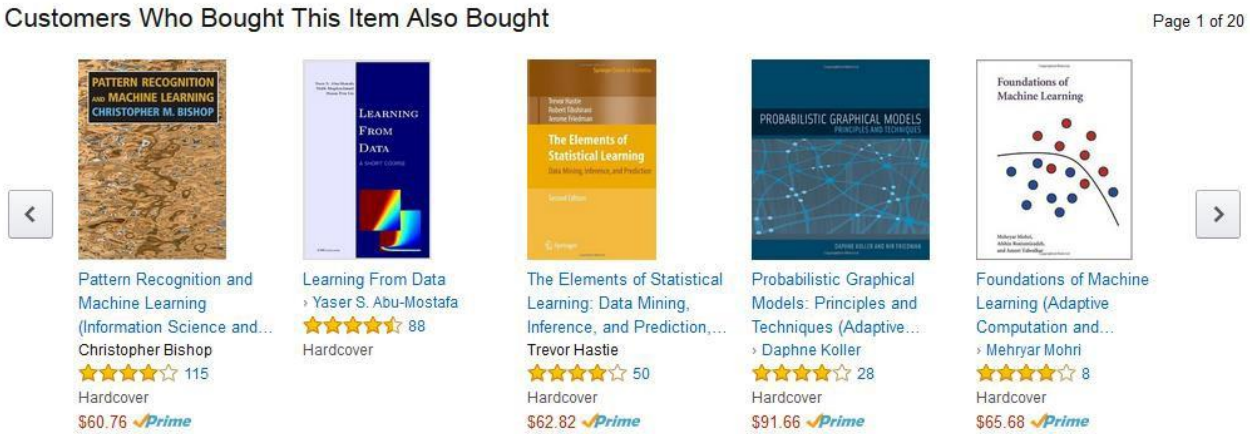
Finding Similar Items

Product Recommendation

- A customer comes to your website looking to buy at item:



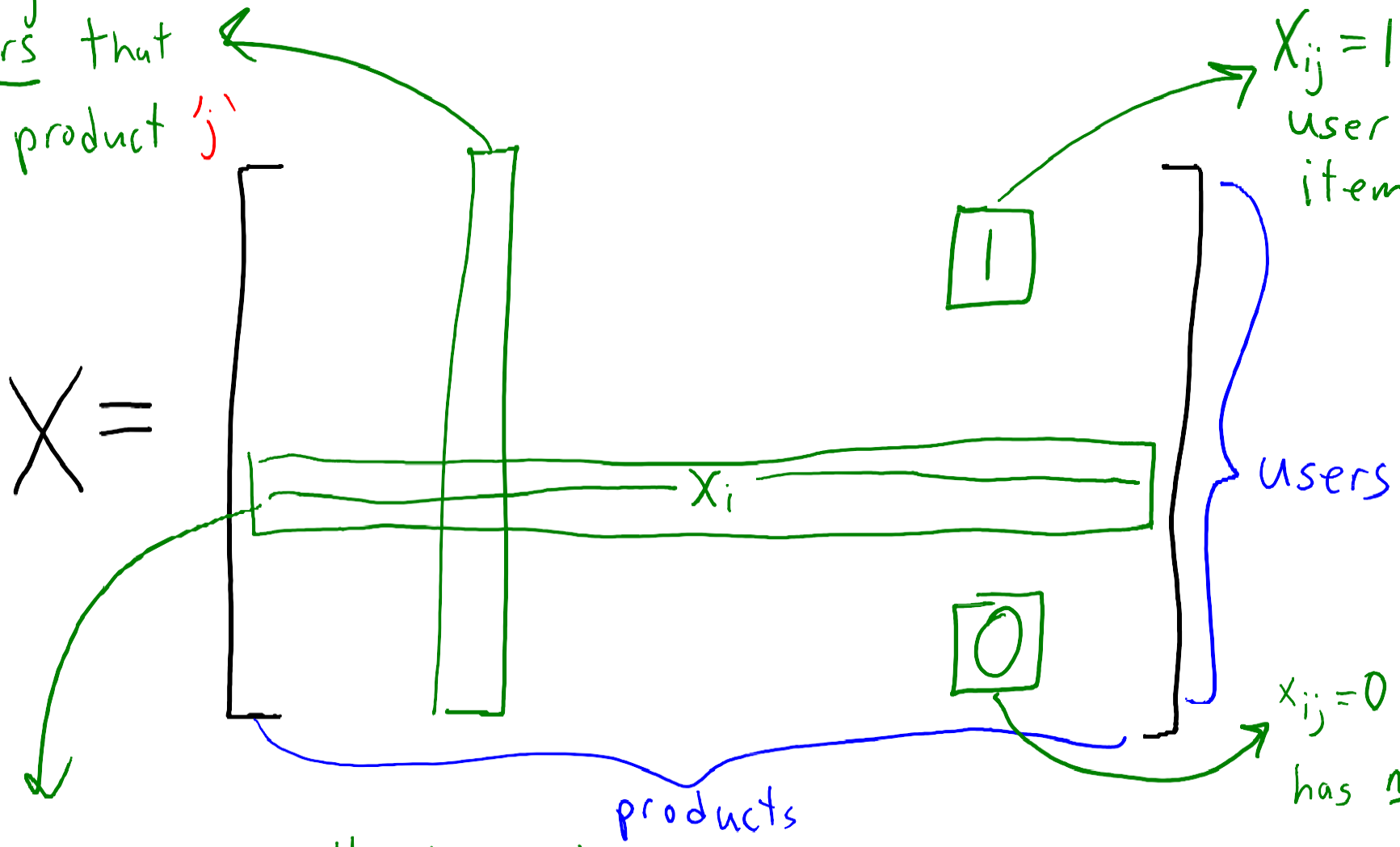
- You want to find similar items that they might also buy:



User-Product Matrix

Column x^j gives all users that bought product 'j'

$X_{ij} = 1$ means user 'i' bought item 'j'!



$x_{ij} = 0$ means user 'i' has not buy item 'j'

Row x_i gives all items bought by user 'i'.

Product Recommendation: Amazon

- Represent each product as a column in a user–product interaction matrix, where rows are users and each cell captures purchase or rating interactions.
- Compare products by measuring similarity across users (two products are considered similar if they tend to be purchased by the same users).
- Return the most similar products as nearest neighbors for each item, building an item-item similarity model.
- At recommendation time, match each product the current user has already purchased to its most similar items, then aggregate those into a final recommendation list.

- Preprocessing step:
  - Normalize each product column before computing similarity.
  - Normalization adjusts for product popularity-items purchased by many users naturally accumulate more signal.
  - This prevents highly popular products from dominating similarity scores.
  - It ensures similarity reflects shared purchasing patterns across users rather than sheer volume of purchases.

Product Recommendation: Amazon

- Consider this user-item matrix:

$$
X = \begin{array}{c} \\ \text{John} \\ \text{Paul} \\ \text{George} \\ \text{Ringo} \\ \text{Yoko} \end{array}
\begin{array}{c}
\begin{array}{cccccc} \text{Product 1} & \text{Product 2} & \text{Product 3} & \text{Product 4} & \text{Product 5} & \text{Product 6} \end{array} \\
\begin{bmatrix}
1 & 1 & 1 & 1 & 0 & 1 \\
1 & 0 & 1 & 0 & 1 & 0 \\
1 & 0 & 1 & 0 & 1 & 1 \\
1 & 0 & 1 & 0 & 1 & 1 \\
1 & 1 & 0 & 1 & 0 & 0
\end{bmatrix}
\end{array}
$$

- Product 1 is most like Product 3 (bought by lots of people).
- Product 2 is most like Product 4 (also bought by John and Yoko).
- Product 3 is equally like Products 1, 5, and 6.
  - Does not consider that Product 1 is more popular than 5 and 6.

Product Recommendation: Amazon

- Consider this user-item matrix (normalized):

$$
X = \begin{array}{c} \\ \text{John} \\ \text{Paul} \\ \text{George} \\ \text{Ringo} \\ \text{Yoko} \end{array}
\begin{array}{c}
\begin{array}{cccccc} \text{Product 1} & \text{Product 2} & \text{Product 3} & \text{Product 4} & \text{Product 5} & \text{Product 6} \end{array} \\
\begin{bmatrix}
1/\sqrt{5} & 1/\sqrt{2} & 1/\sqrt{4} & 1/\sqrt{2} & 0 & 1/\sqrt{3} \\
1/\sqrt{5} & 0 & 1/\sqrt{4} & 0 & 1/\sqrt{3} & 0 \\
1/\sqrt{5} & 0 & 1/\sqrt{4} & 0 & 1/\sqrt{3} & 1/\sqrt{3} \\
1/\sqrt{5} & 0 & 1/\sqrt{4} & 0 & 1/\sqrt{3} & 1/\sqrt{3} \\
1/\sqrt{5} & 1/\sqrt{2} & 0 & 1/\sqrt{2} & 0 & 0
\end{bmatrix}
\end{array}
$$

- Each column represents a product.
- Each column has been normalized to unit length.
- Similarity is computed using cosine similarity (dot product after normalization).
  - Product 1 is most similar to Product 3 (both purchased by many of the same users).
  - Product 2 is most similar to Product 4 (both purchased by John and Yoko).
  - Product 3 is most similar to Product 1 (mutual nearest neighbors).

- **Normalization removes the effect of product popularity.**
- **Similarity depends on *shared purchasing patterns*, not total number of buyers.**
- **This prevents popular products from dominating recommendations.**

http://www.cs.umd.edu/~samir/498/Amazon-Recommendations.pdf

Cost of Finding Nearest Neighbors

- With ‘n’ users and ‘d’ products, finding KNNs cost is O(nd).
  - Not feasible if ‘n’ and ‘d’ are in the millions.
- It’s faster if the user-product matrix is sparse.
  - But data set is still enormous in the Amazon example.
- Closest point problem exists in:
  - KNN classification.
  - K-means clustering.
  - Density-based clustering.
  - Amazon product recommendation.

Recommender Systems

Recommender Systems

- Aim to increase user engagement and maximize revenue by suggesting relevant items to users.
- Online platforms sell items and collect user feedback (e.g., ratings, clicks to improve recommendations.
- Scenarios
  - **Item-based**: Recommend items like a given item
    - Example: “Customers who bought this item also bought…” (Amazon)
  - **User-based**: Recommend items tailored to a specific user
    - Example: Personalized homepage recommendations (Amazon, Netflix)
  - **Hybrid**: Combine user-based and item-based approaches for personalized item recommendations.
- Data:
  - Users typically interact with only a small subset of available items.
  - User/item interaction matrix is highly sparse.
  - It must predict missing ratings/preferences to generate recommendations.

Types of Recommender Systems

- Two Types:
  - Content-Based Filtering: Uses side information (features) about items and/or users.
    - Recommends items similar to those a user has previously liked.
    - Example: Pandora recommending music based on song attributes.
  - Collaborative Filtering: Uses user-item interaction data (e.g., ratings, clicks).
    - Recommends items based on similarities between users or items.

- Supervised Learning in Recommendation:
  - Recommender systems can also be formulated as a supervised learning problem:
  - Extract features x_i describing users and items.
  - Train a model to predict a rating or preference y_i given x_i
  - Apply the trained model to predict preferences for new user-item pairs.

Content Filtering

- Suggests items to a user based on the attributes of items and the user's past preferences. [Focuses on what]
  - Personalized to each user independently
  - Does not require data from other users
- Steps:
  - Item Representation: Each item (e.g., movie, product) is represented by a set of features.
  - User Profile Construction: By aggregating features of items the user has liked, rated highly, or interacted with.
  - Matching Process: Compute similarity between the user profile and candidate items using similarity measures (e.g., cosine similarity,).
  - Recommend items whose feature vectors are most similar to the user's profile.

Content-Based Recommender

```python
import pandas as pd

metadata = pd.read_csv('movies_metadata.csv', low_memory=False)

# Print the first three rows
metadata.head(3)
```

	adult	belongs_to_collection	budget	genres	homepage	id	imdb_id	original_language	original_title	overview	...	release_d
0	False	{'id': 10194, 'name': 'Toy Story Collection', ...	30000000	[{'id': 16, 'name': 'Animation'}, {'id': 35, '...	http://toystory.disney.com/toy-story	862	tt0114709	en	Toy Story	Led by Woody, Andy's toys live happily in his ...	...	1995-10
1	False	NaN	65000000	[{'id': 12, 'name': 'Adventure'}, {'id': 14, '...	NaN	8844	tt0113497	en	Jumanji	When siblings Judy and Peter discover an encha...	...	1995-12
2	False	{'id': 119050, 'name': 'Grumpy Old Men Collect...	0	[{'id': 10749, 'name': 'Romance'}, {'id': 35, ...	NaN	15602	tt0113228	en	Grumpier Old Men	A family wedding reignites the ancient feud be...	...	1995-12

```python
# Calculate C
C = metadata['vote_average'].mean()

# Calculate the minimum number of votes required to be in the chart, m
m = metadata['vote_count'].quantile(0.90)

# Filter out all qualified movies into a new DataFrame
q_movies = metadata.copy().loc[metadata['vote_count'] >= m]
q_movies.shape
```

```
(4555, 24)
```

```python
# Function that computes the weighted rating of each movie
def weighted_rating(x, m=m, C=C):
    v = x['vote_count']
    R = x['vote_average']
    # Calculation based on the IMDB formula
    return (v/(v+m) * R) + (m/(m+v) * C)

# Define a new feature 'score' and calculate its value with `weighted_rating()`
q_movies['score'] = q_movies.apply(weighted_rating, axis=1)
#Sort movies based on score calculated above
q_movies = q_movies.sort_values('score', ascending=False)

#Print the top 15 movies
q_movies[['title', 'vote_count', 'vote_average', 'score']].head(15)
```

	title	vote_count	vote_average	score
314	The Shawshank Redemption	8358.0	8.5	8.445869
834	The Godfather	6024.0	8.5	8.425439
10309	Dilwale Dulhania Le Jayenge	661.0	9.1	8.421453
12481	The Dark Knight	12269.0	8.3	8.265477
2843	Fight Club	9678.0	8.3	8.256385
292	Pulp Fiction	8670.0	8.3	8.251406

```python
#Print plot overviews of the first 5 movies.
metadata['overview'].head(3)
```

```
0    Led by Woody, Andy's toys live happily in his ...
1    When siblings Judy and Peter discover an encha...
2    A family wedding reignites the ancient feud be...
Name: overview, dtype: object
```

Collaborative Filtering

- Collaborative Filtering
  - Does not rely on side information (features) about users or items.
  - Uses only user–item interaction data (e.g., ratings, clicks, purchases).
  - Example: Netflix movie recommendations.
- User preferences are correlated: If Alice and Bob both like item X, and Alice also likes item Y, then Bob is more likely to like item Y.
- Learning Perspective
  - Observe a user-item rating matrix $Y=[y_{ij}]$, where y_{ij} denotes the rating of user i for item j.
  - Matrix is typically sparse.
  - Learns patterns directly from this interaction matrix.

- Limitation: Performs poorly for:
  - New users (no prior ratings)
  - New items (no interaction history)

Collaborative Filtering Problem

- Collaborative filtering is ‘filling in’ the user-item matrix:

$$
Y = \underset{\text{user}}{\left\{\begin{array}{c}\\ \\ \\ \\ \end{array}\right.}
\underbrace{\begin{bmatrix}
? & 4 & 3 & 2 & 3 & 3 \\
2 & 1 & ? & 5 & ? & 5 \\
? & 1 & ? & 5 & 5 & 5 \\
2 & 3 & 3 & ? & ? & ?
\end{bmatrix}}_{\text{movie}}
$$

"Green Lantern"

"Ryan Reynolds"

- We have some ratings available with values {1,2,3,4,5}.
- We want to predict ratings “?” by looking at available ratings.

Collaborative Filtering Problem

$$Y = \begin{array}{c} \text{user} \\ \\ \\ \\ \end{array} \begin{bmatrix} ? & 4 & 3 & 2 & 3 & 3 \\ 2 & 1 & ? & 5 & ? & 5 \\ ? & 1 & ? & 5 & 5 & 5 \\ 2 & 3 & 3 & ? & ? & ? \end{bmatrix}$$

movie

"Green Lantern"

"Ryan Reynolds"

Similar user

Similar movie

- What rating would “Ryan Reynolds” give to “Green Lantern”?
  - Why is this not completely crazy?
  - We may have similar users and movies.

- Two Types of Collaborative filtering methods:
  - Neighborhood
  - Latent Factor

Neighborhood Method

- Identify users with similar movie preferences using a defined similarity metric (e.g., cosine similarity or Pearson correlation).
- Select the top-K most similar users (neighbors).
- Recommend movies that these neighbors have rated highly but the target user has not yet watched.

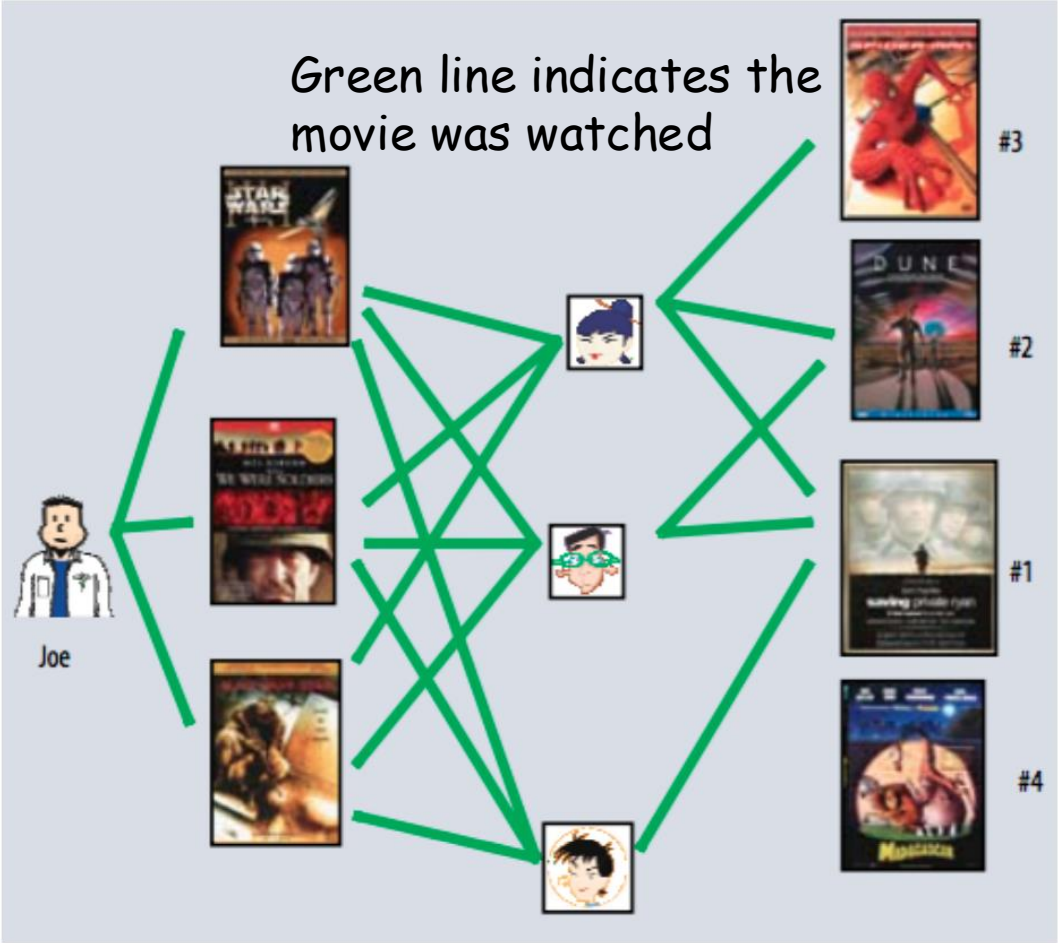


Figure from Koren et al. (2009)

Latent Factor Method

- Assume that both users and movies can be represented as vectors in a shared low-dimensional latent space that captures underlying features (e.g., genre affinity, tone preference, complexity).
  - Learn latent representations for users and movies using techniques (such as matrix factorization).
  - Represent each user and each movie as vectors in this latent feature space.
  - Compute the affinity between a user and a movie (e.g., via dot product or cosine similarity).
  - Recommend movies whose latent vectors are closest to the user's vector, indicating high predicted preference.

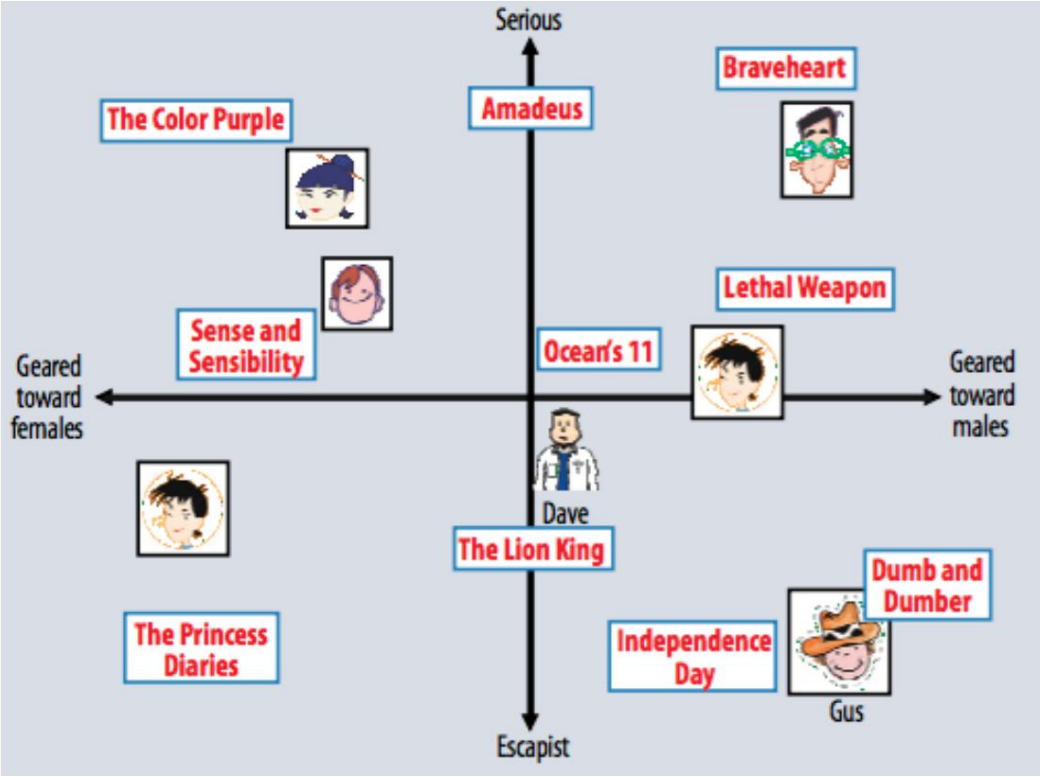


Figure from Koren et al. (2009)

Matrix Factorization (MF)

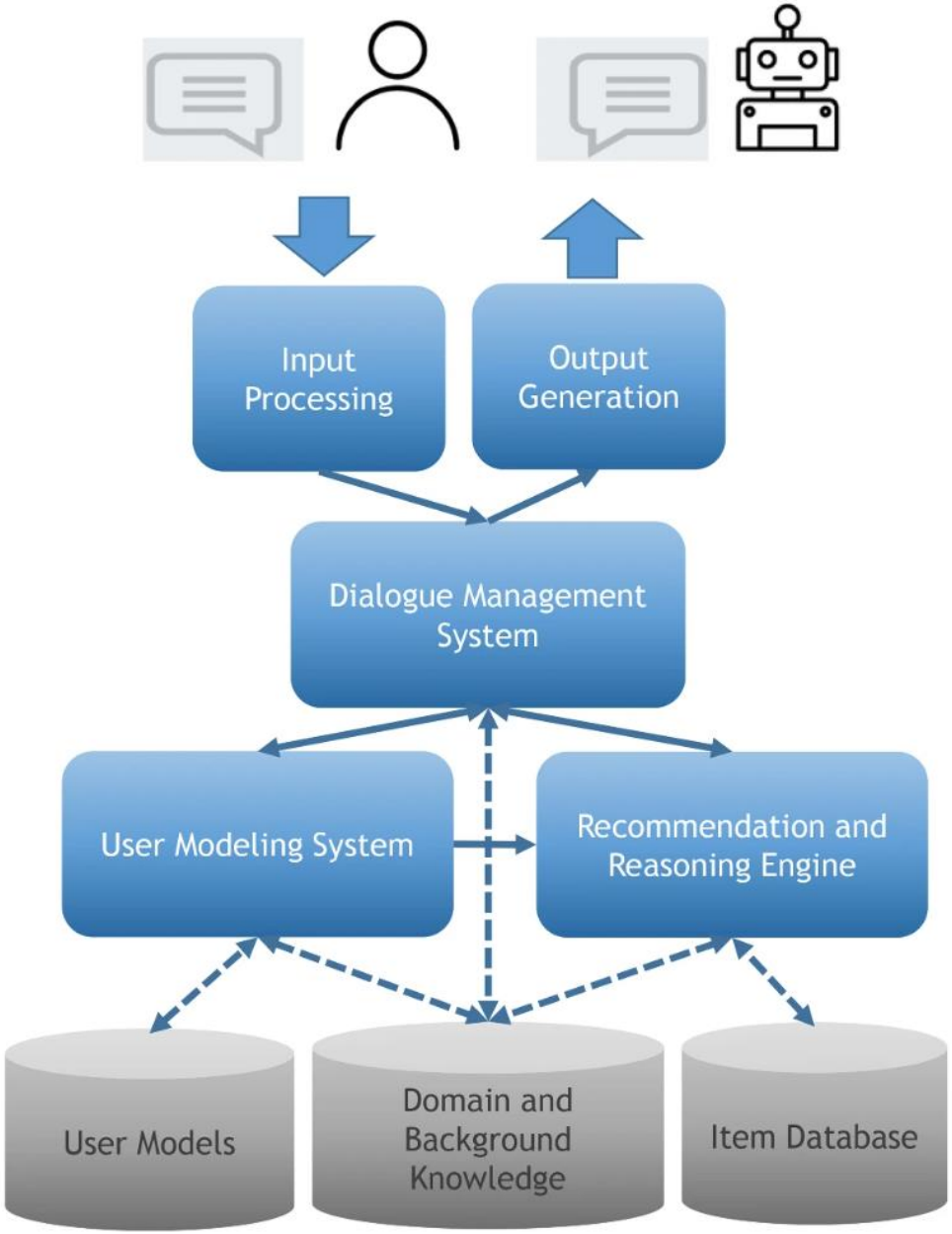
- Collaborative filtering using MF is an efficient and effective approach.
- MF Defines:
  - A model that represents users and items as latent factor vectors.
  - An obj. function that minimizes prediction error (typically squared error with regularization).
  - Parameters are optimized using Stochastic Gradient Descent.
- Classes of Matrix Factorization methods:
  - Unconstrained Matrix Factorization: Learns latent factors without additional structural constraints.
  - Singular Value Decomposition (SVD) adapted to sparse data.
  - Non-negative Matrix Factorization: Constrains latent factors to be non-negative, often improving interpretability.
- Implicit feedback examples:
  - Facebook has a “Like” button but no explicit “Dislike” button.
  - Pinterest interactions (e.g., pins, saves) represent implicit positive feedback.

Issues

- Issues:
  - Diversity: How different are the recommendations?
  - Persistence: How long should recommendations last?
  - Trust: Tell user why you made a recommendation.
    - Quora gives explanations for recommendations.
  - Social recommendation: What did your friends watch?
  - Freshness: people tend to get more excited about new/surprising things.

Conversational recommender systems

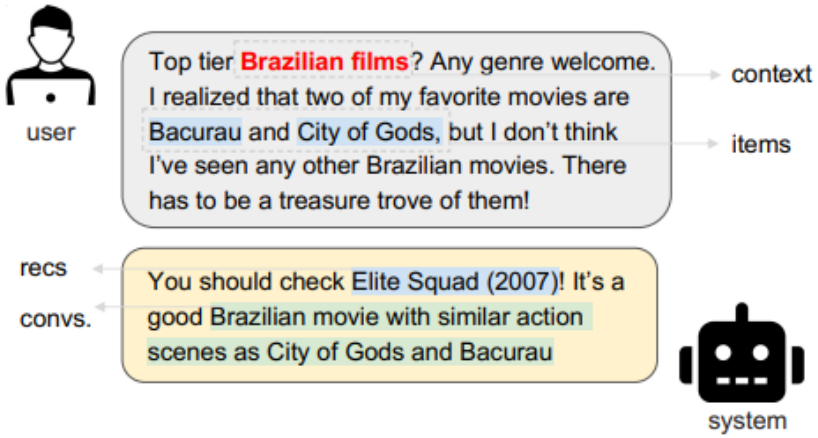
- CRS aim to interactively support online users in their information search and decision-making processes in an intuitive way.



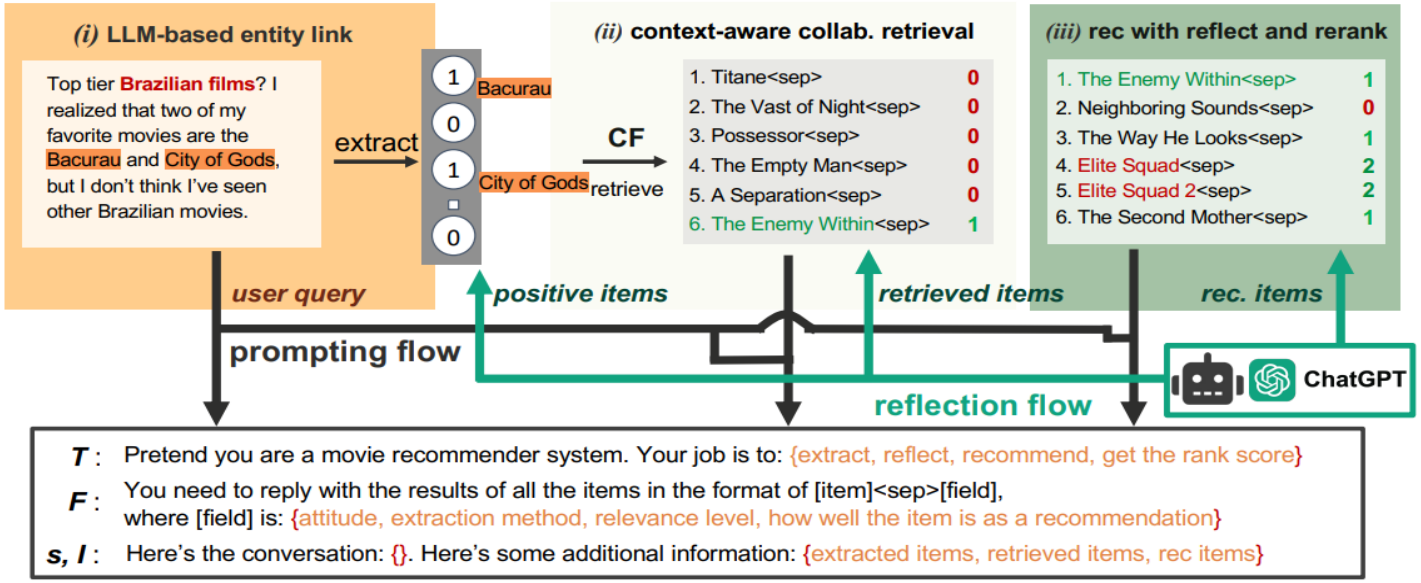
https://link.springer.com/article/10.1007/s10462-022-10229-x

Recommendation System: Netflix

- Conversational recommender systems (CRS) provide personalized item suggestions by iteratively interacting with users through natural language conversations.



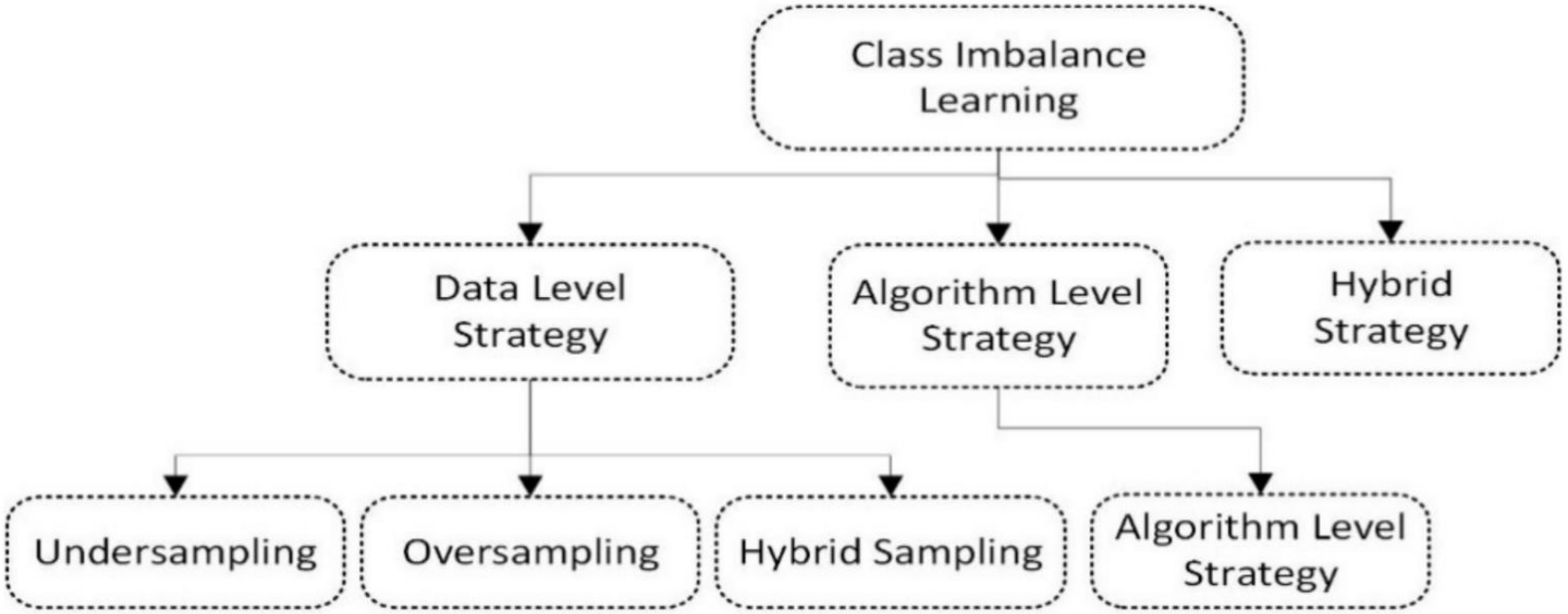
- Collaborative retrieval-augmented LLM for conversational recommendation.
- Main components:
  1. LLM-based entity linking
  2. Collaborative retrieval with context-aware reflection
  3. Recommendation with reflection and re-ranking



https://dl.acm.org/doi/10.1145/3696410.3714908

Class Imbalance

Categorization of class imbalance

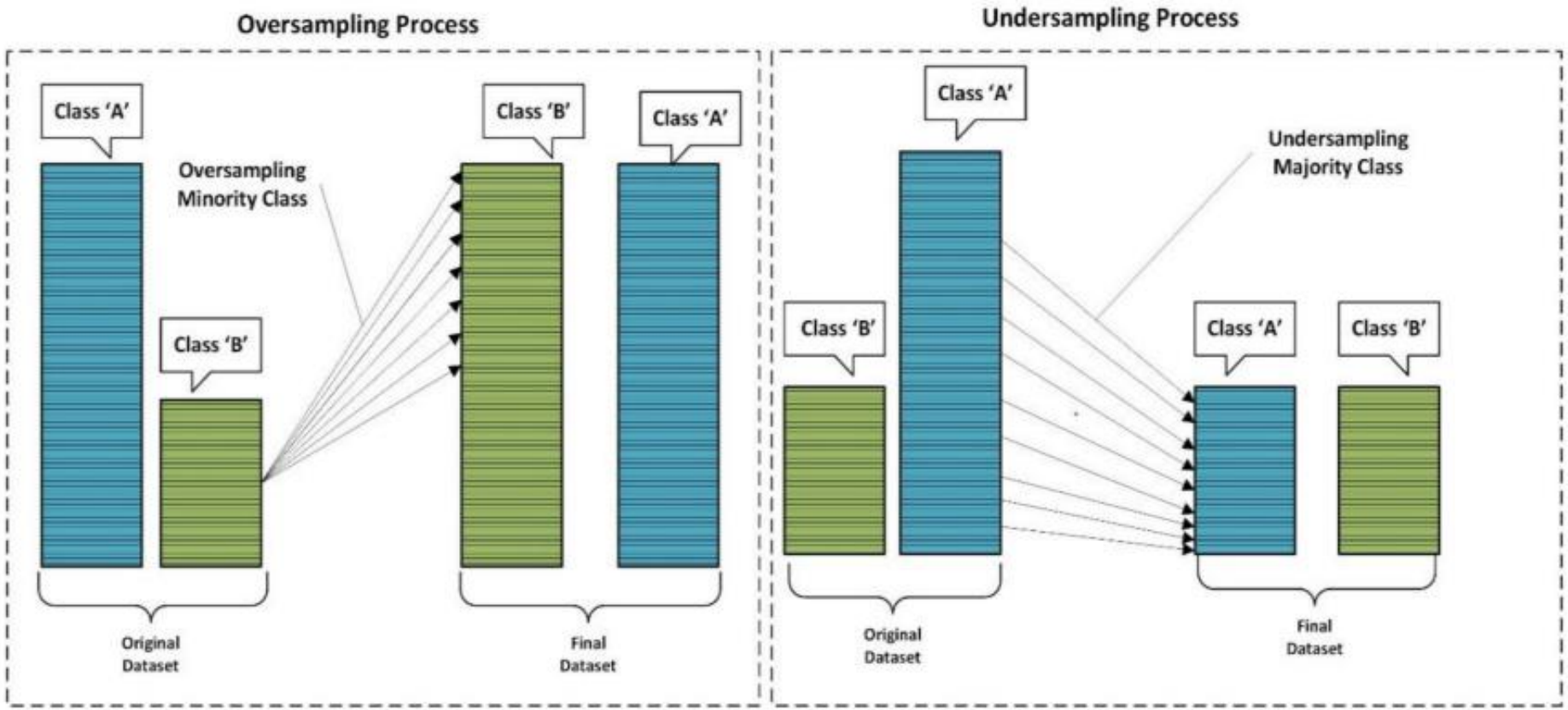


https://doi.org/10.3390/healthcare10071293

Sampling

- Sampling is a statistical process in which a subset of observations is selected from a larger population for analysis.
  - A predetermined number of samples is drawn according to a defined sampling strategy.
- In classification problems with imbalanced classes, sampling techniques are used to adjust the class distribution of the dataset.
- Two common strategies:
  - Oversampling: Increase the number of instances in the minority class.
    - Common techniques: SMOTE & ADASYN
  - Undersampling: Reduce the number of instances in the majority class.
    - Common techniques: Random undersampling & Cluster-based undersampling

Undersampling and oversampling



https://doi.org/10.3390/healthcare10071293

Examples of Class Imbalance

SL	Name	Data Types	Default Task	Attribute Types	#Instances	Class Distribution	#Attributes	Imbalance Ratio
1	Breast Cancer	Multivariate	Classification	Categorical	286	**0**:201,**1**: 85	9	2.36
2	Breast Cancer Wisconsin (Original)	Multivariate	Classification	Integer	699	**0**: 458, **1**:241	10	1.9
3	Breast Cancer Wisconsin (Prognostic)	Multivariate	Classification, Regression	Real	198	**0**:151, **1**:47	34	3.21
4	Breast Cancer Wisconsin (Diagnostic)	Multivariate	Classification	Real	569	**0**:357, **1**:212	32	1.69
5	Heart Disease	Multivariate	Classification	Categorical, Integer, Real	303	**0**:164,**1**:55,2:36,**3**:35,**4**:13	75	–
6	Hepatitis	Multivariate	Classification	Categorical, Integer, Real	155	**0**:133, **1**:32	19	4.15
7	Pima Indians Diabetes Database	Multivariate	Classification	Integer	768	**0**: 500, **1**:268	8	1.9
8	Liver Disorders	Multivariate	Classification	Categorical, Integer, Real	345	**0**:145,**1**:200	7	1.37
9	Lung Cancer	Multivariate	Classification	Integer	32	**0**:23, **1**:9	56	2.55
10	SPECT Heart	Multivariate	Classification	Categorical	267	**0**:55,**1**: 212	22	3.85
11	SPECTF Heart	Multivariate	Classification	Integer	267	**0**:55,**1**:212	44	3.85
12	Thyroid Disease	Multivariate, Domain-Theory	Classification	Categorical, Real	7200	**1**:166, **2**:368, 3:6666	21	–
13	Breast Tissue	Multivariate	Classification	Real	106	**Car**:21 **Fad**:15 **Mas**:8, **Gla**:16, **Con**:14, **Adi**:22	10	–
14	Fertility	Multivariate	Classification, Regression	Real	100	**N**:88, **O**:12	10	7.33
15	Diabetic Retinopathy Debrecen Dataset	Multivariate	Classification	Integer, Real	1151	**0**:540, **1**:611	20	1.131

Imbalance Ratio = (No. of samples in Majority Class)/(No. of samples in Minority Class)

https://doi.org/10.3390/healthcare10071293

Degree of Class Imbalance

Class Imbalance Degree	Proportion of Minority Class
Extreme	<1% of the dataset
Moderate	1–20% of the dataset
Mild	20–40% of the dataset

Dataset	#Instances	#Attributes	Class	IR	Minority Class (%)	Degree of Imbalanced

https://doi.org/10.3390/healthcare10071293

Synthetic Minority Oversampling Technique (SMOTE)

- Causes of Class Imbalance
  - Rare events (e.g., fraud, disease detection)
  - Difficulty or high cost of collecting minority-class data

- Focuses on the under-represented (minority) class in an imbalanced dataset.
- Increases the number of minority class samples in a balanced manner.
- Generates new synthetic instances based on existing minority class examples.
  - Creates new synthetic points in feature space rather than simply duplicating existing ones.

SMOTE

- For each minority class instance:
  - It identifies its k nearest neighbors within the minority class.
  - It generates new synthetic examples by interpolating between the instance and one of its neighbors.

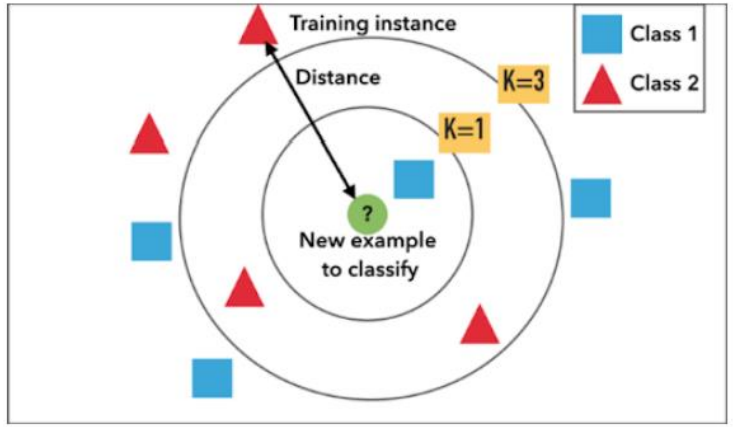
- The new sample is created by combining feature values of the original instance and its neighbor along the line segment connecting them.

- SMOTE takes the entire dataset as input but increases the number of samples only for the minority class, thereby adjusting the class distribution.

https://link.springer.com/article/10.1186/1471-2105-14-106

SMOTE

- SMOTE algorithm:
  - For each minority class sample, find its *k* nearest minority neighbors.
  - Randomly select one of the *k* neighbors.
  - Generate a synthetic sample = {original minority sample + one of its nearest neighbors + gap (0,1)}.
  - Add the generated synthetic samples to the minority class.
  - Combine original and synthetic samples to form the new balanced dataset.



Hussein, A. S., Li, T., Yohannese, C. W., & Bashir, K. (2019). A-SMOTE: A new preprocessing approach for highly imbalanced datasets by improving SMOTE. *International Journal of Computational Intelligence Systems*, *12*(2), 1412-1422.

Example

- In an imbalanced dataset where just 1% of cases have target value A (minority class), and 99% of cases have value B.
  - To increase % of minority cases to 2x the previous percentage, you would enter **200** for **SMOTE percentage** in the component's properties.

	Class 0	Class 1	total
Original dataset	570	178	748
(equivalent to **SMOTE percentage = 0**)	76%	24%	
SMOTE percentage = 100	570	356	926
	62%	38%	
SMOTE percentage = 200	570	534	1,104
	52%	48%	
SMOTE percentage = 300	570	712	1,282
	44%	56%	

Adaptive Synthetic (ADASYN) Sampling

- Extension of SMOTE.
- ADASYN is designed to adaptively generate synthetic samples for the minority class.
- Focuses on regions where minority samples are surrounded by majority samples, thereby shifting the classifier's decision boundary toward difficult areas.
- Process:
  - Synthetic samples are created through linear interpolation between minority class instances (similar to SMOTE).
  - Generates more synthetic samples for minority points that are harder to learn, i.e., those located near the decision boundary between classes.
  - Fewer synthetic samples are generated for minority points located in dense, well-represented interior regions.

He, Haibo, et al. "ADASYN: Adaptive synthetic sampling approach for imbalanced learning." *2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence)*. IEEE, 2008.

ADASYN Process

- Sample generation technique is similar to SMOTE
  - But this time for each minority class data example x_i, $g_i = r_i \times g$ interpolated samples are generated
    - where g is the total number of synthetic data samples needed
    - r_i is proportional to the number of majority class samples between the k nearest neighbours of x_i
      - It is highest near the class boundaries.

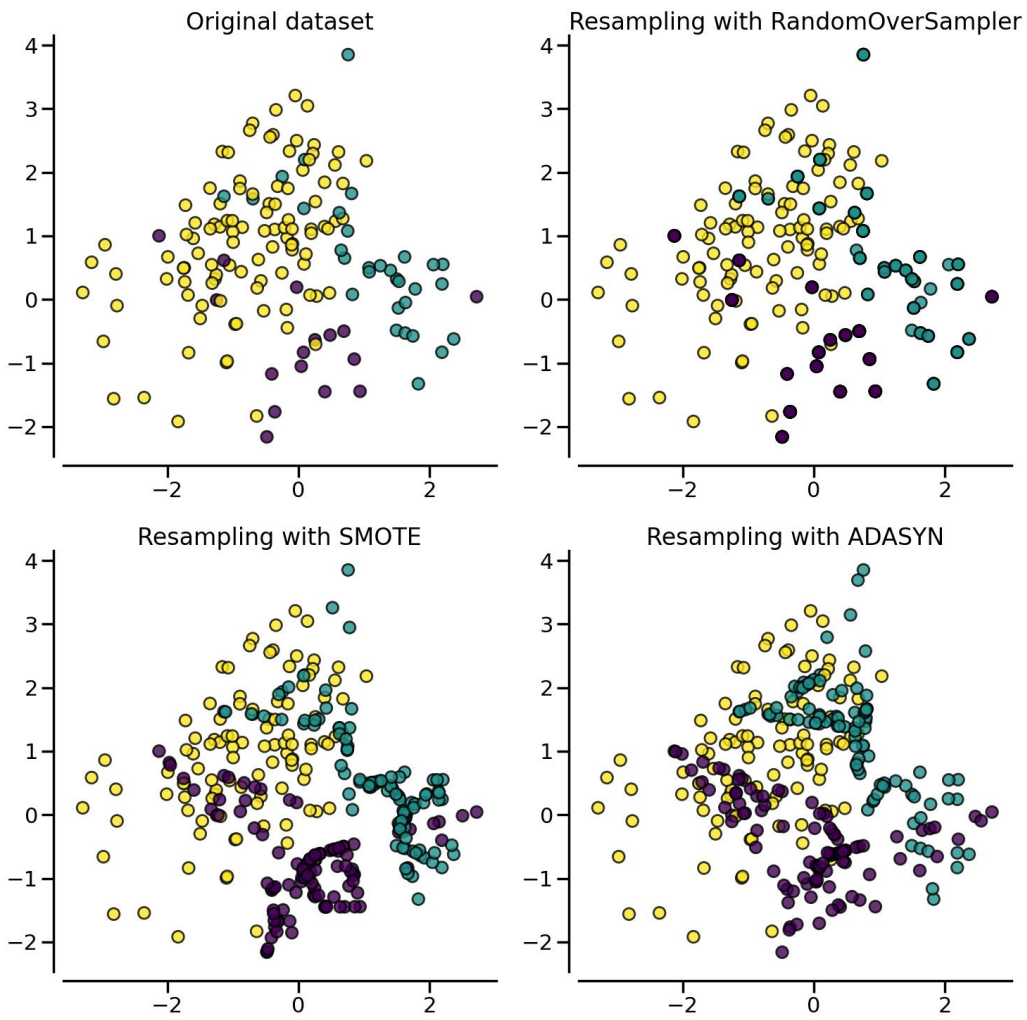
ADASYN (extension of SMOTE)

- ADASYN
  - Shifts the classifier's decision boundary toward regions that are difficult to learn.
  - Improves learning performance on imbalanced datasets.
- Generates synthetic minority class samples to reduce the bias introduced by class imbalance.
- Characteristics
  - Dynamic adjustment of sample generation weights
  - Adaptive learning procedure
  - Focus on hard-to-learn minority instances
- Generation Process:
  - ADASYN uses the local density distribution of the data to determine how many synthetic samples should be generated for each minority instance.
  - Minority samples surrounded by many majority samples (i.e., near the class boundary) receive more synthetic samples, while well-represented minority samples receive fewer.

SMOTE vs ADASYN

```
from imblearn import FunctionSampler  # to use a idendity sampler
from imblearn.over_sampling import ADASYN, SMOTE

X, y = create_dataset(n_samples=150, weights=(0.1, 0.2, 0.7))
fig, axs = plt.subplots(nrows=2, ncols=2, figsize=(15, 15))
samplers = [
    FunctionSampler(),
    RandomOverSampler(random_state=0),
    SMOTE(random_state=0),
    ADASYN(random_state=0),
]
for ax, sampler in zip(axs.ravel(), samplers):
    title = "Original dataset" if isinstance(sampler, FunctionSampler) else None
    plot_resampling(X, y, sampler, ax, title=title)
fig.tight_layout()
```



- SMOTE will not make any distinction.
- Focus on samples which are difficult to classify with a nearest-neighbors rule.

https://imbalanced-learn.org/stable/auto_examples/over-sampling/plot_comparison_over_sampling.html#sphx-glr-auto-examples-over-sampling-plot-comparison-over-sampling-py

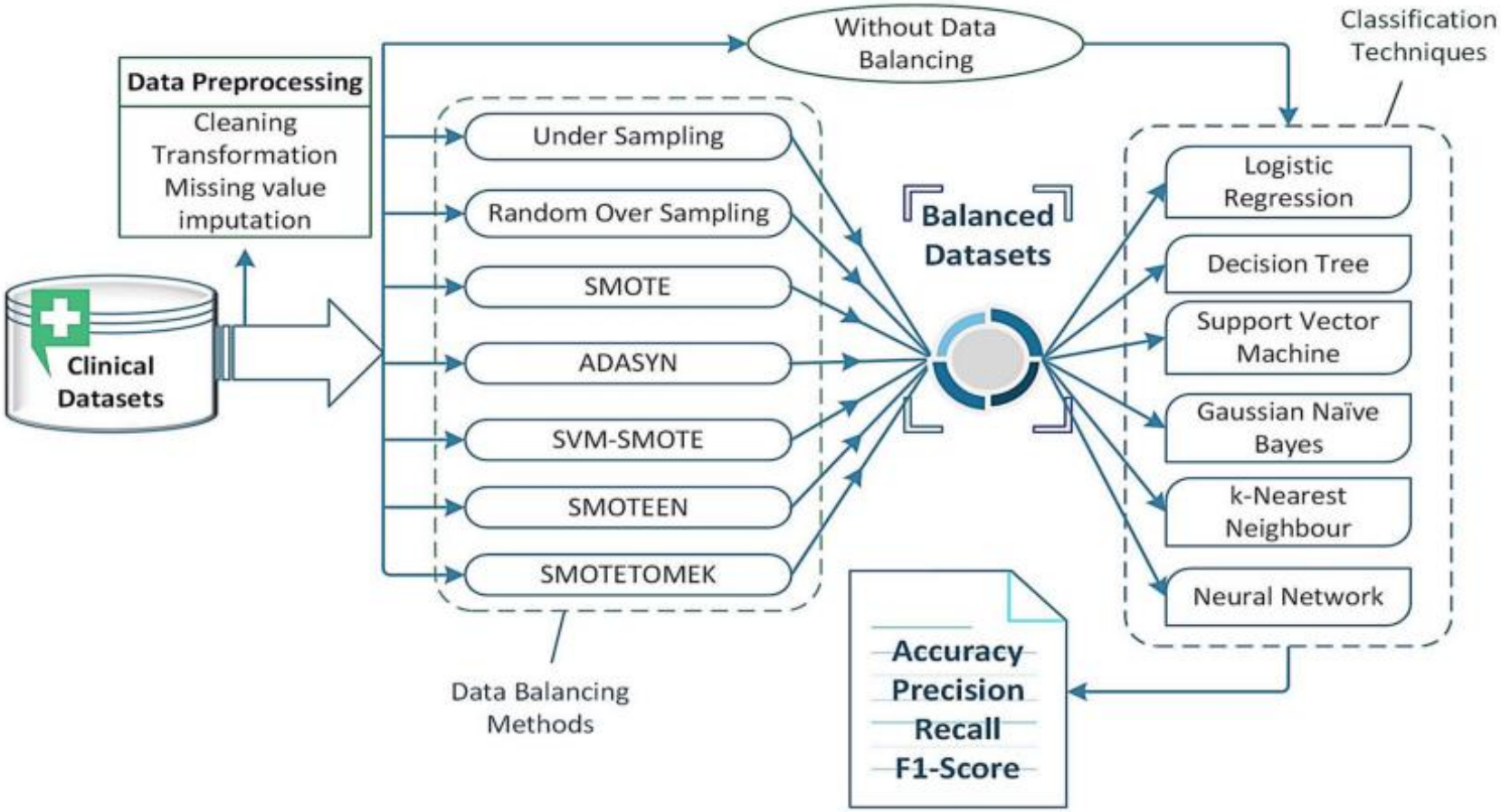
Disadvantage

ADASYN assumes that the space between minority class samples also belongs to the minority class.

This assumption may not hold when the data distribution is complex or not linearly separable.

As a result, synthetic samples generated near class boundaries may overlap with the majority class, potentially increasing noise and reducing classification performance.

Evaluation of classifiers using SMOTE/ADASYN



https://doi.org/10.3390/healthcare10071293

Summary

- Outliers = data points significantly different from the majority
- Types: Global outliers / Local outliers
- Detection Approaches: Graphical, Distance-based, Cluster-based
- Challenges: Hard to define “unusual”

- Finding Similar Items & Recommendations
  - User–Item Matrix is highly sparse
- Similarity-based methods:
  - Item-based (Amazon)
  - User-based (Netflix)

Summary

- Collaborative Filtering:
  - Neighborhood methods
  - Latent Factor / Matrix Factorization
- Class Imbalance Problem
  - Impacts classifier bias
- Sampling Techniques:
  - Oversampling: Increase minority samples
    - SMOTE: Interpolates between minority neighbors
    - ADASYN: Focuses on hard-to-learn boundary regions, Adaptive sample generation
  - Undersampling
    - Reduce majority samples
    - Random or cluster-based